

Limits Drill Sheet

First Calculus Practice: Substitution, Algebra, Factoring, and One-Sided
Limits

300 progressively harder exercises: 50 per topic

Purpose

This drill packet gives first-time calculus students repeated practice recognizing the form of a limit, choosing the correct algebraic move, and deciding when a two-sided limit exists. Each topic has fifty problems in five levels.

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Topic 1

Basic Plug-and-Chug Limits

Learning target

Evaluate limits of polynomial and simple algebraic expressions by direct substitution.

Worked example

Problem. $\lim_{x \rightarrow 2} (3x^2 - 4x + 1)$

Move. Substitute $x = 2$ because polynomials are continuous.

Result. $3(2)^2 - 4(2) + 1 = 5$.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow -3} x - 4$

6. $\lim_{x \rightarrow 2} 2x + 1$

2. $\lim_{x \rightarrow -2} 2x - 3$

7. $\lim_{x \rightarrow 3} 3x + 2$

3. $\lim_{x \rightarrow -1} 3x - 2$

8. $\lim_{x \rightarrow 4} 4x + 3$

4. $\lim_{x \rightarrow 0} 4x - 1$

9. $\lim_{x \rightarrow -4} x + 4$

5. $\lim_{x \rightarrow 1} x$

10. $\lim_{x \rightarrow 5} 2x + 5$

Level II: routine variations

1. $\lim_{x \rightarrow -3} x^2 + 2x - 3$

6. $\lim_{x \rightarrow 2} 3x^2 + 2x + 2$

2. $\lim_{x \rightarrow -2} 2x^2 + x - 2$

7. $\lim_{x \rightarrow 3} x^2 + x + 3$

3. $\lim_{x \rightarrow -1} 3x^2 - 1$

8. $\lim_{x \rightarrow 4} 2x^2 - 3$

4. $\lim_{x \rightarrow 0} x^2 - x$

9. $\lim_{x \rightarrow -4} 3x^2 - x - 2$

5. $\lim_{x \rightarrow 1} 2x^2 - 2x + 1$

10. $\lim_{x \rightarrow 5} x^2 - 2x - 1$

Level III: disguised or prepared forms

1. $\lim_{x \rightarrow -3} x^2 + 7x + 9$

2. $\lim_{x \rightarrow -2} x^2 + 6x + 3$

3. $\lim_{x \rightarrow -1} x^2 + 5x - 1$

4. $\lim_{x \rightarrow 0} x^2 + 4x - 3$

5. $\lim_{x \rightarrow 1} x^2 + 3x + 1$

6. $\lim_{x \rightarrow 2} x^2 - 3x + 3$

7. $\lim_{x \rightarrow 3} x^2 - 4x + 7$

8. $\lim_{x \rightarrow 4} x^2 - 5x + 13$

9. $\lim_{x \rightarrow -4} x^2 + 12x + 16$

10. $\lim_{x \rightarrow 5} x^2 - 5x + 24$

Level IV: multi-stage execution

1. $\lim_{x \rightarrow -3} x^3 + 2x - 7$

2. $\lim_{x \rightarrow -2} 2x^3 - x^2 + 3x - 7$

3. $\lim_{x \rightarrow -1} x^3 - 2x^2 + 4x - 7$

4. $\lim_{x \rightarrow 0} 2x^3 - 3x^2 + 5x - 7$

5. $\lim_{x \rightarrow 1} x^3 - 4x^2 + 2x - 7$

6. $\lim_{x \rightarrow 2} 2x^3 + 3x - 7$

7. $\lim_{x \rightarrow 3} x^3 - x^2 + 4x - 7$

8. $\lim_{x \rightarrow 4} 2x^3 - 2x^2 + 5x - 7$

9. $\lim_{x \rightarrow -4} x^3 - 3x^2 + 2x - 7$

10. $\lim_{x \rightarrow 5} 2x^3 - 4x^2 + 3x - 7$

Level V: independent mastery

1. $\lim_{x \rightarrow -3} x^3 + 5x^2 + 14x + 7$

2. $\lim_{x \rightarrow -2} 2x^3 + 10x^2 + 28x + 15$

3. $\lim_{x \rightarrow -1} 3x^3 + 15x^2 + 42x + 23$

4. $\lim_{x \rightarrow 0} x^3 + 2x^2 + 20x + 7$

5. $\lim_{x \rightarrow 1} 2x^3 + 11x^2 + 26x + 19$

6. $\lim_{x \rightarrow 2} 3x^3 + 16x^2 + 40x + 27$

7. $\lim_{x \rightarrow 3} x^3 + 3x^2 + 18x + 11$

8. $\lim_{x \rightarrow 4} 2x^3 + 8x^2 + 32x + 19$

9. $\lim_{x \rightarrow -4} 3x^3 + 17x^2 + 38x + 31$

10. $\lim_{x \rightarrow 5} x^3 + 4x^2 + 16x + 15$

Topic 2

Rational Expressions by Direct Substitution

Learning target

Evaluate rational limits when the denominator is not zero at the target value.

Worked example

Problem. $\lim_{x \rightarrow -1} \frac{x^2 + 3x + 5}{x + 4}$

Move. Substitute because the denominator becomes 3, not 0.

Result. $\frac{1-3+5}{3} = 1$.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow -3} \frac{x - 3}{x + 10}$

2. $\lim_{x \rightarrow -2} \frac{2x - 2}{x + 11}$

3. $\lim_{x \rightarrow -1} \frac{3x - 1}{x + 12}$

4. $\lim_{x \rightarrow 0} \frac{4x}{x + 13}$

5. $\lim_{x \rightarrow 1} \frac{x + 1}{x + 14}$

6. $\lim_{x \rightarrow 2} \frac{2x + 2}{x + 15}$

7. $\lim_{x \rightarrow 3} \frac{3x + 3}{x + 16}$

8. $\lim_{x \rightarrow 4} \frac{4x + 4}{x + 17}$

9. $\lim_{x \rightarrow -4} \frac{x + 5}{x + 18}$

10. $\lim_{x \rightarrow 5} \frac{2x + 6}{x + 19}$

Level II: routine variations

1. $\lim_{x \rightarrow -3} \frac{x^2 - 2x + 3}{x - 1}$

2. $\lim_{x \rightarrow -2} \frac{2x^2 - x + 3}{x - 3}$

$$3. \lim_{x \rightarrow -1} \frac{3x^2 + 3}{x - 5}$$

$$4. \lim_{x \rightarrow 0} \frac{x^2 + x + 3}{x - 7}$$

$$5. \lim_{x \rightarrow 1} \frac{2x^2 + 2x + 3}{x - 9}$$

$$6. \lim_{x \rightarrow 2} \frac{3x^2 - 2x + 3}{x - 11}$$

$$7. \lim_{x \rightarrow 3} \frac{x^2 - x + 3}{x - 13}$$

$$8. \lim_{x \rightarrow 4} \frac{2x^2 + 3}{x - 15}$$

$$9. \lim_{x \rightarrow -4} \frac{3x^2 + x + 3}{x - 8}$$

$$10. \lim_{x \rightarrow 5} \frac{x^2 + 2x + 3}{x - 18}$$

Level III: disguised or prepared forms

$$1. \lim_{x \rightarrow -3} \frac{x^2 + 3x + 2}{x^2 + x - 56}$$

$$2. \lim_{x \rightarrow -2} \frac{x^2 + 3x + 2}{x^2 + x - 72}$$

$$3. \lim_{x \rightarrow -1} \frac{x^2 + 3x}{x^2 + x - 90}$$

$$4. \lim_{x \rightarrow 0} \frac{x^2 + 6x + 8}{x^2 + x - 110}$$

$$5. \lim_{x \rightarrow 1} \frac{x^2 + 2x + 1}{x^2 + 5x - 84}$$

$$6. \lim_{x \rightarrow 2} \frac{x^2 + 2x}{x^2 - 64}$$

$$7. \lim_{x \rightarrow 3} \frac{x^2 + 5x + 6}{x^2 - 81}$$

$$8. \lim_{x \rightarrow 4} \frac{x^2 + 5x + 4}{x^2 - 100}$$

$$9. \lim_{x \rightarrow -4} \frac{x^2 + x}{x^2 + 4x - 77}$$

$$10. \lim_{x \rightarrow 5} \frac{x^2 + 4x + 4}{x^2 + 4x - 96}$$

Level IV: multi-stage execution

$$1. \lim_{x \rightarrow -3} \frac{\frac{x+1}{x+8}}{\frac{x-2}{x+11}}$$

$$2. \lim_{x \rightarrow -2} \frac{\frac{2x+1}{x+9}}{\frac{x-2}{x+12}}$$

$$3. \lim_{x \rightarrow -1} \frac{\frac{3x+1}{x+10}}{\frac{x-2}{x+13}}$$

$$4. \lim_{x \rightarrow 0} \frac{\frac{x+1}{x+11}}{\frac{x-2}{x+14}}$$

$$5. \lim_{x \rightarrow 1} \frac{\frac{2x+1}{x+12}}{\frac{x-2}{x+15}}$$

$$6. \lim_{x \rightarrow 2} \frac{\frac{3x+1}{x+13}}{\frac{x-2}{x+16}}$$

$$7. \lim_{x \rightarrow 3} \frac{\frac{x+1}{x+14}}{\frac{x-2}{x+11}}$$

$$8. \lim_{x \rightarrow 4} \frac{\frac{2x+1}{x+15}}{\frac{x-2}{x+12}}$$

$$9. \lim_{x \rightarrow -4} \frac{\frac{3x+1}{x+16}}{\frac{x-2}{x+13}}$$

$$10. \lim_{x \rightarrow 5} \frac{\frac{x+1}{x+17}}{\frac{x-2}{x+14}}$$

Level V: independent mastery

$$1. \lim_{x \rightarrow -3} \frac{x^3 - 2x - 5}{x^2 + 8x + 30}$$

$$2. \lim_{x \rightarrow -2} \frac{2x^3 - 2x - 4}{x^2 + 9x + 31}$$

3. $\lim_{x \rightarrow -1} \frac{3x^3 - 2x - 3}{x^2 + 10x + 32}$

4. $\lim_{x \rightarrow 0} \frac{x^3 - 2x - 2}{x^2 + 11x + 33}$

5. $\lim_{x \rightarrow 1} \frac{2x^3 - 2x - 1}{x^2 + 8x + 34}$

6. $\lim_{x \rightarrow 2} \frac{3x^3 - 2x}{x^2 + 9x + 35}$

7. $\lim_{x \rightarrow 3} \frac{x^3 - 2x + 1}{x^2 + 10x + 36}$

8. $\lim_{x \rightarrow 4} \frac{2x^3 - 2x + 2}{x^2 + 11x + 37}$

9. $\lim_{x \rightarrow -4} \frac{3x^3 - 2x + 3}{x^2 + 8x + 38}$

10. $\lim_{x \rightarrow 5} \frac{x^3 - 2x + 4}{x^2 + 9x + 39}$

Topic 3

Factoring and Cancellation Limits

Learning target

Handle 0/0 rational limits by factoring completely, cancelling common factors, and then substituting.

Worked example

Problem. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Move. Factor $x^2 - 9 = (x - 3)(x + 3)$, cancel the common factor, then substitute.

Result. 6.

Drill

Level I: direct recognition

1. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

2. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$

3. $\lim_{x \rightarrow -5} \frac{x^2 - 25}{x + 5}$

4. $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$

5. $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$

6. $\lim_{x \rightarrow -3} \frac{2x^2 + 7x + 3}{x + 3}$

7. $\lim_{x \rightarrow 4} \frac{3x^2 - 11x - 4}{x - 4}$

8. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

9. $\lim_{x \rightarrow -3} \frac{x^3 + 27}{x + 3}$

10. $\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$

Level II: routine variations

1. $\lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 9x + 18}{x^2 + 4x - 21}$

2. $\lim_{x \rightarrow 4} \frac{2x^3 - x^2 - 32x + 16}{x^2 + 4x - 32}$

3. $\lim_{x \rightarrow -5} \frac{3x^3 - 75x}{x^2 + 14x + 45}$

4. $\lim_{x \rightarrow -2} \frac{x^3 + 6x^2 + 11x + 6}{x^2 + 12x + 20}$

$$5. \lim_{x \rightarrow 4} \frac{2x^3 - 26x - 24}{x^2 + 3x - 28}$$

$$6. \lim_{x \rightarrow -3} \frac{6x^3 + 27x^2 + 30x + 9}{x^2 + 11x + 24}$$

$$7. \lim_{x \rightarrow 4} \frac{3x^3 + x^2 - 48x - 16}{x^2 + 5x - 36}$$

$$8. \lim_{x \rightarrow 2} \frac{2x^4 + 5x^3 - 16x - 40}{x^2 + 8x - 20}$$

$$9. \lim_{x \rightarrow -3} \frac{3x^4 + 6x^3 + 81x + 162}{x^2 + 10x + 21}$$

$$10. \lim_{x \rightarrow 2} \frac{x^5 + 7x^4 - 16x - 112}{x^2 + 6x - 16}$$

Level III: disguised or prepared forms

$$1. \lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 6x + 9}{x^3 + 6x^2 - 9x - 54}$$

$$2. \lim_{x \rightarrow 4} \frac{2x^3 - 7x^2 - 7x + 12}{x^3 + 5x^2 - 16x - 80}$$

$$3. \lim_{x \rightarrow -5} \frac{3x^3 + 16x^2 + 2x - 15}{x^3 + 4x^2 - 25x - 100}$$

$$4. \lim_{x \rightarrow -2} \frac{4x^3 + 9x^2 - x - 6}{x^3 + 8x^2 + 21x + 18}$$

$$5. \lim_{x \rightarrow 4} \frac{x^3 - 3x^2 - 7x + 12}{x^3 + x^2 - 14x - 24}$$

$$6. \lim_{x \rightarrow -3} \frac{2x^3 + 7x^2 - 9}{2x^3 + 19x^2 + 45x + 18}$$

$$7. \lim_{x \rightarrow 4} \frac{3x^3 - 11x^2 - 7x + 12}{3x^3 + 4x^2 - 59x - 20}$$

$$8. \lim_{x \rightarrow 2} \frac{4x^3 - 7x^2 - 5x + 6}{x^4 + 4x^3 - 8x - 32}$$

$$9. \lim_{x \rightarrow -3} \frac{x^3 + 4x^2 - 9}{x^4 + 3x^3 + 27x + 81}$$

$$10. \lim_{x \rightarrow 2} \frac{2x^3 - 3x^2 - 5x + 6}{x^5 + 2x^4 - 16x - 32}$$

Level IV: multi-stage execution

$$1. \lim_{x \rightarrow 3} \frac{x^3 + x^2 - 9x - 9}{x^3 - x^2 - 5x - 3}$$

$$2. \lim_{x \rightarrow 4} \frac{x^3 + 2x^2 - 16x - 32}{x^3 - x^2 - 11x - 4}$$

$$3. \lim_{x \rightarrow -5} \frac{x^3 + 3x^2 - 25x - 75}{x^3 + 9x^2 + 21x + 5}$$

$$4. \lim_{x \rightarrow -2} \frac{x^3 + 9x^2 + 26x + 24}{x^3 + 7x^2 + 11x + 2}$$

$$5. \lim_{x \rightarrow 4} \frac{x^3 + 4x^2 - 17x - 60}{x^3 - 2x^2 - 7x - 4}$$

$$6. \lim_{x \rightarrow -3} \frac{2x^3 + 9x^2 + 10x + 3}{x^3 + 6x^2 + 10x + 3}$$

$$7. \lim_{x \rightarrow 4} \frac{3x^3 - 5x^2 - 26x - 8}{x^3 - 15x - 4}$$

$$8. \lim_{x \rightarrow 2} \frac{x^4 + 3x^3 - 8x - 24}{x^3 + 3x^2 - 9x - 2}$$

$$9. \lim_{x \rightarrow -3} \frac{x^4 + 4x^3 + 27x + 108}{x^3 + 5x^2 + 7x + 3}$$

$$10. \lim_{x \rightarrow 2} \frac{x^5 + 5x^4 - 16x - 80}{x^3 + x^2 - 5x - 2}$$

Level V: independent mastery

$$1. \lim_{x \rightarrow 3} \frac{x^3 - 2x^2 - 5x + 6}{x^3 + 5x^2 - 9x - 45}$$

$$2. \lim_{x \rightarrow 4} \frac{x^3 - 3x^2 - 10x + 24}{x^3 + 6x^2 - 16x - 96}$$

$$3. \lim_{x \rightarrow -5} \frac{x^3 + 6x^2 - 7x - 60}{x^3 + 7x^2 - 25x - 175}$$

$$4. \lim_{x \rightarrow -2} \frac{x^3 + 6x^2 + 3x - 10}{x^3 + 13x^2 + 46x + 48}$$

$$5. \lim_{x \rightarrow 4} \frac{x^3 - 4x^2 - 4x + 16}{x^3 + 8x^2 - 21x - 108}$$

$$6. \lim_{x \rightarrow -3} \frac{x^3 + 3x^2 - 9x - 27}{2x^3 + 27x^2 + 73x + 30}$$

7. $\lim_{x \rightarrow 4} \frac{x^3 - x^2 - 16x + 16}{3x^3 + 4x^2 - 59x - 20}$

8. $\lim_{x \rightarrow 2} \frac{x^3 + x^2 - 16x + 20}{x^4 + 6x^3 - 8x - 48}$

9. $\lim_{x \rightarrow -3} \frac{x^3 + 2x^2 - 9x - 18}{x^4 + 7x^3 + 27x + 189}$

10. $\lim_{x \rightarrow 2} \frac{x^3 - 7x + 6}{x^5 + 8x^4 - 16x - 128}$

Topic 4

Left and Right Limits

Learning target

Evaluate left-hand and right-hand limits of piecewise functions by using the rule on the correct side of the target.

Worked example

Problem. If $f(x) = \begin{cases} x+1, & x < 2 \\ 3x, & x > 2 \end{cases}$, find both one-sided limits at $x = 2$.

Result. $\lim_{x \rightarrow 2^-} f(x) = 3$ and $\lim_{x \rightarrow 2^+} f(x) = 6$.

Drill

Level I: direct recognition

1. $f(x) = \begin{cases} x, & x < -3 \\ 2x, & x > -3 \end{cases}$; find $\lim_{x \rightarrow -3^-} f(x)$ and $\lim_{x \rightarrow -3^+} f(x)$
2. $f(x) = \begin{cases} x+1, & x < -2 \\ 2x-1, & x > -2 \end{cases}$; find $\lim_{x \rightarrow -2^-} f(x)$ and $\lim_{x \rightarrow -2^+} f(x)$
3. $f(x) = \begin{cases} x+2, & x < -1 \\ 2x-2, & x > -1 \end{cases}$; find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$
4. $f(x) = \begin{cases} x+3, & x < 0 \\ 2x-3, & x > 0 \end{cases}$; find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$
5. $f(x) = \begin{cases} x+4, & x < 1 \\ 2x-4, & x > 1 \end{cases}$; find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$
6. $f(x) = \begin{cases} x+5, & x < 2 \\ 2x-5, & x > 2 \end{cases}$; find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$
7. $f(x) = \begin{cases} x+6, & x < 3 \\ 2x-6, & x > 3 \end{cases}$; find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$
8. $f(x) = \begin{cases} x+7, & x < 4 \\ 2x-7, & x > 4 \end{cases}$; find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$
9. $f(x) = \begin{cases} x+8, & x < -4 \\ 2x-8, & x > -4 \end{cases}$; find $\lim_{x \rightarrow -4^-} f(x)$ and $\lim_{x \rightarrow -4^+} f(x)$
10. $f(x) = \begin{cases} x+9, & x < 5 \\ 2x-9, & x > 5 \end{cases}$; find $\lim_{x \rightarrow 5^-} f(x)$ and $\lim_{x \rightarrow 5^+} f(x)$

Level II: routine variations

1. $f(x) = \begin{cases} x^2, & x < -3 \\ -x, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
2. $f(x) = \begin{cases} x^2 + 1, & x < -2 \\ 3 - x, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$
3. $f(x) = \begin{cases} x^2 + 2, & x < -1 \\ 6 - x, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
4. $f(x) = \begin{cases} x^2 + 3, & x < 0 \\ 9 - x, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
5. $f(x) = \begin{cases} x^2 + 4, & x < 1 \\ 12 - x, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
6. $f(x) = \begin{cases} x^2 + 5, & x < 2 \\ 15 - x, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
7. $f(x) = \begin{cases} x^2 + 6, & x < 3 \\ 18 - x, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
8. $f(x) = \begin{cases} x^2 + 7, & x < 4 \\ 21 - x, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
9. $f(x) = \begin{cases} x^2 + 8, & x < -4 \\ 24 - x, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
10. $f(x) = \begin{cases} x^2 + 9, & x < 5 \\ 27 - x, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Level III: disguised or prepared forms

1. $f(x) = \begin{cases} \frac{x^2-9}{x+3}, & x < -3 \\ x+2, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
2. $f(x) = \begin{cases} \frac{x^2-4}{x+2}, & x < -2 \\ 2x+2, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$
3. $f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x < -1 \\ 3x+2, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
4. $f(x) = \begin{cases} \frac{x^2+x}{x}, & x < 0 \\ x+2, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
5. $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 2x+2, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
6. $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x < 2 \\ 3x+2, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
7. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x < 3 \\ x+2, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
8. $f(x) = \begin{cases} \frac{x^2-16}{x-4}, & x < 4 \\ 2x+2, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
9. $f(x) = \begin{cases} \frac{x^2-16}{x+4}, & x < -4 \\ 3x+2, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
10. $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x < 5 \\ x+2, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Level IV: multi-stage execution

1. $f(x) = \begin{cases} x^2 + 6x + 9, & x < -3 \\ \frac{x^2+2x+1}{x+9}, & x > -3 \end{cases}$; find $\lim_{x \rightarrow -3^-} f(x)$ and $\lim_{x \rightarrow -3^+} f(x)$
2. $f(x) = \begin{cases} x^2 + 4x + 5, & x < -2 \\ \frac{x^2+x+1}{x+10}, & x > -2 \end{cases}$; find $\lim_{x \rightarrow -2^-} f(x)$ and $\lim_{x \rightarrow -2^+} f(x)$

3. $f(x) = \begin{cases} x^2 + 2x + 3, & x < -1 \\ \frac{x^2+1}{x+11}, & x > -1 \end{cases}$; find $\lim_{x \rightarrow -1^-} f(x)$ and $\lim_{x \rightarrow -1^+} f(x)$
7. $f(x) = \begin{cases} x^2 - 6x + 15, & x < 3 \\ \frac{x^2+1}{x+15}, & x > 3 \end{cases}$; find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$
4. $f(x) = \begin{cases} x^2 + 3, & x < 0 \\ \frac{x^2-x+1}{x+12}, & x > 0 \end{cases}$; find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$
8. $f(x) = \begin{cases} x^2 - 8x + 23, & x < 4 \\ \frac{x^2-x+1}{x+16}, & x > 4 \end{cases}$; find $\lim_{x \rightarrow 4^-} f(x)$ and $\lim_{x \rightarrow 4^+} f(x)$
5. $f(x) = \begin{cases} x^2 - 2x + 5, & x < 1 \\ \frac{x^2+2x+1}{x+13}, & x > 1 \end{cases}$; find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$
9. $f(x) = \begin{cases} x^2 + 8x + 24, & x < -4 \\ \frac{x^2+2x+1}{x+17}, & x > -4 \end{cases}$; find $\lim_{x \rightarrow -4^-} f(x)$ and $\lim_{x \rightarrow -4^+} f(x)$
6. $f(x) = \begin{cases} x^2 - 4x + 9, & x < 2 \\ \frac{x^2+x+1}{x+14}, & x > 2 \end{cases}$; find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$
10. $f(x) = \begin{cases} x^2 - 10x + 34, & x < 5 \\ \frac{x^2+x+1}{x+18}, & x > 5 \end{cases}$; find $\lim_{x \rightarrow 5^-} f(x)$ and $\lim_{x \rightarrow 5^+} f(x)$

Level V: independent mastery

1. $f(x) = \begin{cases} \frac{x^2+4x+3}{x+3}, & x < -3 \\ \frac{2x^2+6x}{x+3}, & x > -3 \end{cases}$; find both one-sided limits at $x = -3$
6. $f(x) = \begin{cases} \frac{x^2+4x-12}{x-2}, & x < 2 \\ \frac{2x^2-9x+10}{x-2}, & x > 2 \end{cases}$; find both one-sided limits at $x = 2$
2. $f(x) = \begin{cases} \frac{x^2+4x+4}{x+2}, & x < -2 \\ \frac{2x^2+3x-2}{x+2}, & x > -2 \end{cases}$; find both one-sided limits at $x = -2$
7. $f(x) = \begin{cases} \frac{x^2+4x-21}{x-3}, & x < 3 \\ \frac{2x^2-12x+18}{x-3}, & x > 3 \end{cases}$; find both one-sided limits at $x = 3$
3. $f(x) = \begin{cases} \frac{x^2+4x+3}{x+1}, & x < -1 \\ \frac{2x^2-2}{x+1}, & x > -1 \end{cases}$; find both one-sided limits at $x = -1$
8. $f(x) = \begin{cases} \frac{x^2+4x-32}{x-4}, & x < 4 \\ \frac{2x^2-15x+28}{x-4}, & x > 4 \end{cases}$; find both one-sided limits at $x = 4$
4. $f(x) = \begin{cases} \frac{x^2+4x}{x}, & x < 0 \\ \frac{2x^2-3x}{x}, & x > 0 \end{cases}$; find both one-sided limits at $x = 0$
9. $f(x) = \begin{cases} \frac{x^2+13x+36}{x+4}, & x < -4 \\ \frac{2x^2-32}{x+4}, & x > -4 \end{cases}$; find both one-sided limits at $x = -4$
5. $f(x) = \begin{cases} \frac{x^2+4x-5}{x-1}, & x < 1 \\ \frac{2x^2-6x+4}{x-1}, & x > 1 \end{cases}$; find both one-sided limits at $x = 1$
10. $f(x) = \begin{cases} \frac{x^2+5x-50}{x-5}, & x < 5 \\ \frac{2x^2-19x+45}{x-5}, & x > 5 \end{cases}$; find both one-sided limits at $x = 5$

Topic 5

One-Sided Limits That Disagree

Learning target

Compare one-sided limits and conclude that the two-sided limit does not exist when they disagree.

Worked example

Problem. If $f(x) = \begin{cases} x, & x < 1 \\ x + 4, & x > 1 \end{cases}$, decide whether $\lim_{x \rightarrow 1} f(x)$ exists.

Result. The left-hand limit is 1 and the right-hand limit is 5, so the two-sided limit does not exist.

Drill

Level I: direct recognition

- | | |
|--|---|
| 1. $f(x) = \begin{cases} x, & x < -3 \\ x + 3, & x > -3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -3} f(x)$ exists | 6. $f(x) = \begin{cases} x + 5, & x < 2 \\ x + 8, & x > 2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 2} f(x)$ exists |
| 2. $f(x) = \begin{cases} x + 1, & x < -2 \\ x + 4, & x > -2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -2} f(x)$ exists | 7. $f(x) = \begin{cases} x + 6, & x < 3 \\ x + 9, & x > 3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 3} f(x)$ exists |
| 3. $f(x) = \begin{cases} x + 2, & x < -1 \\ x + 5, & x > -1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -1} f(x)$ exists | 8. $f(x) = \begin{cases} x + 7, & x < 4 \\ x + 10, & x > 4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 4} f(x)$ exists |
| 4. $f(x) = \begin{cases} x + 3, & x < 0 \\ x + 6, & x > 0 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 0} f(x)$ exists | 9. $f(x) = \begin{cases} x + 8, & x < -4 \\ x + 11, & x > -4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -4} f(x)$ exists |
| 5. $f(x) = \begin{cases} x + 4, & x < 1 \\ x + 7, & x > 1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 1} f(x)$ exists | 10. $f(x) = \begin{cases} x + 9, & x < 5 \\ x + 12, & x > 5 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 5} f(x)$ exists |

Level II: routine variations

1. $f(x) = \begin{cases} 2x, & x < -3 \\ -x, & x > -3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -3$
2. $f(x) = \begin{cases} 2x - 1, & x < -2 \\ 1 - x, & x > -2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -2$
3. $f(x) = \begin{cases} 2x - 2, & x < -1 \\ 2 - x, & x > -1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -1$
4. $f(x) = \begin{cases} 2x - 3, & x < 0 \\ 3 - x, & x > 0 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 0$
5. $f(x) = \begin{cases} 2x - 4, & x < 1 \\ 4 - x, & x > 1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 1$
6. $f(x) = \begin{cases} 2x - 5, & x < 2 \\ 5 - x, & x > 2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 2$
7. $f(x) = \begin{cases} 2x - 6, & x < 3 \\ 6 - x, & x > 3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 3$
8. $f(x) = \begin{cases} 2x - 7, & x < 4 \\ 7 - x, & x > 4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 4$
9. $f(x) = \begin{cases} 2x - 8, & x < -4 \\ 8 - x, & x > -4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -4$
10. $f(x) = \begin{cases} 2x - 9, & x < 5 \\ 9 - x, & x > 5 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 5$

Level III: disguised or prepared forms

1. $f(x) = \begin{cases} \frac{x^2+5x+6}{x+3}, & x < -3 \\ x+5, & x > -3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -3} f(x)$ exists
2. $f(x) = \begin{cases} \frac{x^2+5x+6}{x+2}, & x < -2 \\ x+6, & x > -2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -2} f(x)$ exists
3. $f(x) = \begin{cases} \frac{x^2+5x+4}{x+1}, & x < -1 \\ x+7, & x > -1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -1} f(x)$ exists
4. $f(x) = \begin{cases} \frac{x^2+5x}{x}, & x < 0 \\ x+8, & x > 0 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 0} f(x)$ exists
5. $f(x) = \begin{cases} \frac{x^2+5x-6}{x-1}, & x < 1 \\ x+9, & x > 1 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 1} f(x)$ exists
6. $f(x) = \begin{cases} \frac{x^2+5x-14}{x-2}, & x < 2 \\ x+10, & x > 2 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 2} f(x)$ exists
7. $f(x) = \begin{cases} \frac{x^2+5x-24}{x-3}, & x < 3 \\ x+11, & x > 3 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 3} f(x)$ exists
8. $f(x) = \begin{cases} \frac{x^2+5x-36}{x-4}, & x < 4 \\ x+12, & x > 4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 4} f(x)$ exists
9. $f(x) = \begin{cases} \frac{x^2+14x+40}{x+4}, & x < -4 \\ x+13, & x > -4 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow -4} f(x)$ exists
10. $f(x) = \begin{cases} \frac{x^2+6x-55}{x-5}, & x < 5 \\ x+14, & x > 5 \end{cases}$; find both one-sided limits and decide whether $\lim_{x \rightarrow 5} f(x)$ exists

Level IV: multi-stage execution

1. $f(x) = \begin{cases} x^2 + 6x + 9, & x < -3 \\ x^2 + 6x + 13, & x > -3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -3$
2. $f(x) = \begin{cases} x^2 + 4x + 5, & x < -2 \\ x^2 + 4x + 9, & x > -2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -2$

3. $f(x) = \begin{cases} x^2 + 2x + 3, & x < -1 \\ x^2 + 2x + 7, & x > -1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -1$
4. $f(x) = \begin{cases} x^2 + 3, & x < 0 \\ x^2 + 7, & x > 0 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 0$
5. $f(x) = \begin{cases} x^2 - 2x + 5, & x < 1 \\ x^2 - 2x + 9, & x > 1 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 1$
6. $f(x) = \begin{cases} x^2 - 4x + 9, & x < 2 \\ x^2 - 4x + 13, & x > 2 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 2$
7. $f(x) = \begin{cases} x^2 - 6x + 15, & x < 3 \\ x^2 - 6x + 19, & x > 3 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 3$
8. $f(x) = \begin{cases} x^2 - 8x + 23, & x < 4 \\ x^2 - 8x + 27, & x > 4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 4$
9. $f(x) = \begin{cases} x^2 + 8x + 24, & x < -4 \\ x^2 + 8x + 28, & x > -4 \end{cases}$; find both one-sided limits and the two-sided limit at $x = -4$
10. $f(x) = \begin{cases} x^2 - 10x + 34, & x < 5 \\ x^2 - 10x + 38, & x > 5 \end{cases}$; find both one-sided limits and the two-sided limit at $x = 5$

Level V: independent mastery

1. $f(x) = \begin{cases} \frac{x^2-9}{x+3}, & x < -3 \\ 2x+1, & x > -3 \end{cases}$; find both one-sided limits and decide whether the limit exists
2. $f(x) = \begin{cases} \frac{x^2-4}{x+2}, & x < -2 \\ 2x+2, & x > -2 \end{cases}$; find both one-sided limits and decide whether the limit exists
3. $f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x < -1 \\ 2x+3, & x > -1 \end{cases}$; find both one-sided limits and decide whether the limit exists
4. $f(x) = \begin{cases} \frac{x^2+x}{x}, & x < 0 \\ 2x+4, & x > 0 \end{cases}$; find both one-sided limits and decide whether the limit exists
5. $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 2x+5, & x > 1 \end{cases}$; find both one-sided limits and decide whether the limit exists
6. $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x < 2 \\ 2x+6, & x > 2 \end{cases}$; find both one-sided limits and decide whether the limit exists
7. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x < 3 \\ 2x+7, & x > 3 \end{cases}$; find both one-sided limits and decide whether the limit exists
8. $f(x) = \begin{cases} \frac{x^2-16}{x-4}, & x < 4 \\ 2x+8, & x > 4 \end{cases}$; find both one-sided limits and decide whether the limit exists
9. $f(x) = \begin{cases} \frac{x^2-16}{x+4}, & x < -4 \\ 2x+9, & x > -4 \end{cases}$; find both one-sided limits and decide whether the limit exists
10. $f(x) = \begin{cases} \frac{x^2-25}{x-5}, & x < 5 \\ 2x+10, & x > 5 \end{cases}$; find both one-sided limits and decide whether the limit exists

Topic 6

Limit Algebra Laws and Reciprocals

Learning target

Use sum, difference, product, quotient, power, root, and reciprocal limit laws from known component limits.

Worked example

Problem. Suppose $\lim_{x \rightarrow c} f(x) = 2$ and $\lim_{x \rightarrow c} g(x) = -3$. Find $\lim_{x \rightarrow c} (f(x)g(x) + 4)$.

Result. $2(-3) + 4 = -2$.

Drill

Level I: direct recognition

- Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} f(x) + g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} f(x) - g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} 3f(x) + 2g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} f(x)g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$.
- Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{1}{f(x)}$.
- Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} f(x)^2 + g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} h(x) - f(x)g(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x) + h(x)}{g(x)}$.
- Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{g(x)h(x)}{f(x)}$.

Level II: routine variations

- Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} 2f(x) - 3g(x) + h(x)$.
- Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x)}{g(x) - f(x)}$.
- Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and

- $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} (f(x) + g(x))^2$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{1}{g(x)^2}$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} f(x)^2 - g(x)^2$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x)g(x) + h(x)}{2}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x) - g(x)}{f(x) + 1}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x) + 2}{h(x) - 1}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 + h(x)}{g(x)}$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{1}{f(x)g(x)}$.

Level III: disguised or prepared forms

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \sqrt{h(x) + 5}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{f(x)^2 + g(x)^2}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{\sqrt{h(x) + 12}}{g(x)}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} f(x)^3 - 2h(x)$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{4}{h(x)}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x) - g(x)}{h(x) + 2}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{h(x)^2}{f(x) - g(x)}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x) + g(x) + h(x)}{3}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} (2f(x) - g(x))h(x)$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{1}{h(x) - f(x)}$.

Level IV: multi-stage execution

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 + g(x)^2}{h(x)}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{h(x)^2 + f(x)^2}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{g(x)}{f(x)^2 + 1}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{h(x) - 2g(x)}{f(x) + 3}$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{f(x)h(x)}{g(x)^2}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{1}{\sqrt{h(x) + 9}}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{f(x) + g(x) + 10}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{f(x)^2 - g(x)}{h(x) - g(x)}$.

9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \left(\frac{f(x)}{g(x)} \right)^2$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{h(x) + 1}{f(x)g(x) - 2}$.

Level V: independent mastery

1. Suppose $\lim_{x \rightarrow c} f(x) = 2$, $\lim_{x \rightarrow c} g(x) = -3$, and $\lim_{x \rightarrow c} h(x) = 4$. Find $\lim_{x \rightarrow c} \frac{2}{f(x)} + \frac{3}{g(x)}$.
2. Suppose $\lim_{x \rightarrow c} f(x) = -1$, $\lim_{x \rightarrow c} g(x) = 5$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} + \frac{g(x)}{h(x)}$.
3. Suppose $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = -4$. Find $\lim_{x \rightarrow c} \frac{h(x)}{f(x)} - \frac{f(x)}{h(x)}$.
4. Suppose $\lim_{x \rightarrow c} f(x) = 0$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 7$. Find $\lim_{x \rightarrow c} \frac{f(x)^2}{g(x)} + h(x)$.
5. Suppose $\lim_{x \rightarrow c} f(x) = 5$, $\lim_{x \rightarrow c} g(x) = 1$, and $\lim_{x \rightarrow c} h(x) = -1$. Find $\lim_{x \rightarrow c} \frac{1}{f(x) + g(x)}$.
6. Suppose $\lim_{x \rightarrow c} f(x) = -4$, $\lim_{x \rightarrow c} g(x) = 3$, and $\lim_{x \rightarrow c} h(x) = 6$. Find $\lim_{x \rightarrow c} \frac{f(x) + g(x)}{f(x) - g(x)}$.
7. Suppose $\lim_{x \rightarrow c} f(x) = 6$, $\lim_{x \rightarrow c} g(x) = -1$, and $\lim_{x \rightarrow c} h(x) = 2$. Find $\lim_{x \rightarrow c} \sqrt{h(x) + f(x)^2}$.
8. Suppose $\lim_{x \rightarrow c} f(x) = 1$, $\lim_{x \rightarrow c} g(x) = 4$, and $\lim_{x \rightarrow c} h(x) = -3$. Find $\lim_{x \rightarrow c} \frac{h(x)^2 - f(x)^2}{g(x)}$.
9. Suppose $\lim_{x \rightarrow c} f(x) = -2$, $\lim_{x \rightarrow c} g(x) = -5$, and $\lim_{x \rightarrow c} h(x) = 8$. Find $\lim_{x \rightarrow c} \frac{f(x) + 1}{g(x) - 1}$.
10. Suppose $\lim_{x \rightarrow c} f(x) = 7$, $\lim_{x \rightarrow c} g(x) = 2$, and $\lim_{x \rightarrow c} h(x) = 3$. Find $\lim_{x \rightarrow c} \frac{h(x) + g(x)}{f(x)^2 + 2}$.

Before Declaring the Limit Complete

- I checked whether direct substitution is allowed.
- I factored before cancelling in any $0/0$ rational limit.
- I cancelled only common factors, never separate terms.
- I used the left-hand rule for $x \rightarrow a^-$ and the right-hand rule for $x \rightarrow a^+$.
- I concluded DNE only after comparing the two one-sided limits.
- I avoided the quotient and reciprocal laws when the denominator limit is 0.